

SURA, Russian Federation

WMO: 226760

Lat: 63.583N

Lon: 45.633E

Elev: 204

StdP: 14.59

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.5	-27.3	-38.8	0.6	-34.2	-31.6	0.9	-27.1	19.1	19.6	17.5	20.0	1.6	160	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	82.3	65.3	78.5	63.8	74.8	61.9	67.9	77.7	66.0	75.0	63.9	71.8	8.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.9	71.6	62.6	85.6	69.6	60.5	79.4	68.3	32.4	77.9	30.8	75.2	29.3	72.0	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.8	14.1	DB	-37.6	86.6	7.4	3.4	-42.9	89.1	-47.3	91.1	-51.4	93.0	-56.8	95.5	(4)
(5)			WB	-37.7	70.5	7.4	2.9	-43.0	72.6	-47.3	74.3	-51.5	75.9	-56.8	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	6.5	7.8	19.4	32.9	45.3	55.1	61.2	56.2	46.7	35.0	22.0	12.8		
	DBStd	21.92	17.05	16.71	11.34	9.17	9.51	8.41	7.46	7.25	6.63	7.24	12.40	16.22		
	HDD50	6868	1350	1181	948	515	210	40	6	20	139	465	841	1153		
	HDD65	11554	1815	1601	1413	962	612	316	163	286	548	929	1291	1618		
	CDD50	870	0	0	0	3	65	192	354	214	41	1	0	0		
	CDD65	81	0	0	0	0	2	17	47	15	0	0	0	0		
	CDH74	935	0	0	0	1	44	236	499	153	2	0	0	0		
	CDH80	200	0	0	0	0	7	54	106	33	0	0	0	0		
	WSAvg	6.6	6.1	6.4	6.9	6.9	7.4	6.8	6.0	6.1	6.4	7.1	6.6	6.4		
	PrecAvg	21.9	1.5	1.1	1.1	1.2	1.9	2.4	2.6	2.7	2.1	2.2	1.8	1.5		
(15) Precipitation	PrecMax	27.3	3.1	2.0	1.8	2.4	3.1	6.2	5.9	4.9	3.9	4.1	3.0	2.3		
	PrecMin	18.1	0.7	0.2	0.2	0.3	0.6	0.8	0.7	1.3	0.9	0.6	0.6	0.9		
	PrecStd	2.4	0.5	0.4	0.4	0.6	0.7	1.2	1.2	0.9	0.8	0.9	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	36.4	45.1	65.6	80.2	85.9	87.6	84.6	70.3	55.5	43.8	37.8		
		MCWB	33.1	34.8	36.5	50.0	59.9	65.7	68.3	68.6	59.4	49.9	42.7	36.3		
	2%	DB	32.6	33.8	39.3	55.3	72.7	80.4	83.4	78.0	65.0	50.4	40.2	34.6		
		MCWB	31.9	32.9	34.5	44.2	55.7	62.9	66.3	65.1	56.8	47.0	39.1	33.3		
	5%	DB	30.3	31.8	36.5	49.2	68.0	75.8	79.7	73.3	60.9	47.6	37.4	33.4		
		MCWB	29.4	30.8	33.2	40.5	54.3	60.5	64.9	62.7	54.8	45.6	36.7	32.5		
	10%	DB	27.6	29.2	34.4	45.5	63.2	71.3	76.4	69.0	57.3	45.0	34.8	31.9		
		MCWB	26.8	28.2	32.0	39.0	51.3	58.7	63.5	61.0	53.0	43.4	34.0	31.1		
	0.4%	WB	33.6	35.1	38.0	51.6	62.5	68.3	71.5	70.2	61.0	51.5	42.9	36.1		
		MCDB	33.8	35.9	41.6	64.7	72.2	80.9	82.8	80.5	67.5	54.9	43.7	37.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.0	33.0	35.6	45.5	58.4	65.6	68.7	66.8	58.0	47.8	39.3	33.7		
		MCDB	32.5	33.6	38.5	52.8	69.8	76.7	78.6	75.3	63.2	49.6	40.1	34.2		
	5%	WB	29.6	30.9	33.9	42.1	55.1	62.9	67.0	64.1	55.7	45.6	36.7	32.6		
		MCDB	30.3	31.6	35.7	47.8	66.1	72.1	76.0	70.5	59.7	47.4	37.3	33.4		
	10%	WB	26.8	28.3	32.3	39.5	52.2	60.1	65.1	62.0	53.4	43.4	34.1	31.0		
		MCDB	27.6	29.1	34.4	44.5	61.3	68.8	73.5	67.8	57.1	45.0	34.6	32.0		
	5% DB	MDBR	12.7	14.7	17.4	16.6	18.3	18.5	18.8	16.1	13.1	7.6	8.1	10.4		
		MCDBR	9.0	8.0	15.4	23.3	27.6	26.1	25.3	23.9	20.7	11.6	6.7	6.7		
		MCWBR	9.2	7.6	11.6	14.0	13.6	11.5	10.5	11.5	12.7	9.2	6.8	6.5		
		MCWBR	9.3	8.6	10.9	20.4	23.9	22.3	21.5	20.5	17.8	11.2	6.8	6.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.5	8.3	8.7	13.1	13.1	11.4	10.2	11.0	11.9	9.3	6.9	6.5		
	taub		0.231	0.285	0.314	0.351	0.350	0.351	0.392	0.365	0.333	0.307	0.237	0.217		
		taud	1.955	2.060	2.103	2.146	2.433	2.485	2.374	2.428	2.506	2.365	2.034	1.940		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		109	195	241	259	274	276	259	256	243	195	107	48		
		Edh at Noon	10	22	31	37	31	30	33	29	22	16	9	4		
	RadAvg		58	251	677	1219	1521	1652	1623	1121	601	219	61	23		
(44) All-Sky Solar Radiation	RadStd		8	53	101	139	106	166	137	98	69	33	13	3		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page